

Trainings ▾

Preparatory Steps

Automotive Applications

- Remove the fan and water pump drive belt. Refer to Procedure 008-003 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-003.html>).
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13 (</qs3/pubsys2/xml/en/procedures/10/10-013-005.html>).
- Remove the fan and water pump belt tensioner. Refer to Procedure 008-080 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>).



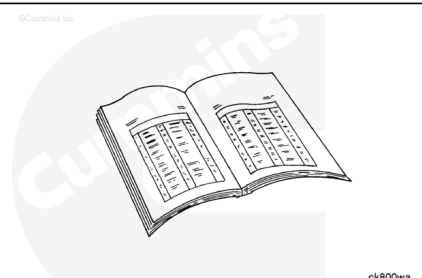
Industrial and Power Generation with Crank-Driven Fan or PTO

- Remove the fan. See equipment manufacturer service information.
- Remove the fan and water pump drive belts. Refer to Procedure 008-003 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-003.html>).
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13 (</qs3/pubsys2/xml/en/procedures/10/10-013-005.html>).
- Remove the fan and water pump belt tensioner. Refer to Procedure 008-080 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>).



Industrial and Power Generation without Crank-Driven Fan or PTO

- Remove the fan. See equipment manufacturer service information.
- Remove the fan and water pump drive belts. Refer to Procedure 008-003 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-003.html>).



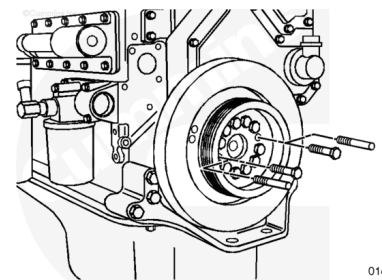
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13 (</qs3/pubsys2/xml/en/procedures/10/10-013-005.html>).
- Remove the fan and water pump tensioner. Refer to Procedure 008-080 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>).

Remove

Automotive Applications

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



01c00148

⚠ CAUTION ⚠

Do not use a hammer or a screwdriver to remove the vibration damper(s). These tools can damage the vibration damper(s).

Remove two of the vibration damper(s) and crankshaft pulley retaining capscrews.

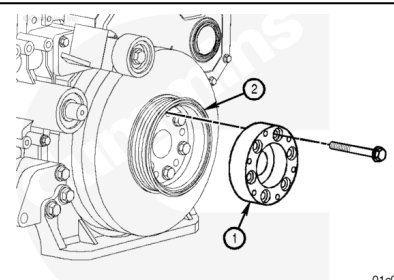
Install two guide studs into the holes.

Remove the remaining 10 capscrews.

Remove the pulley and vibration damper(s).

Industrial and Power Generation with Crank-Driven Fan or PTO

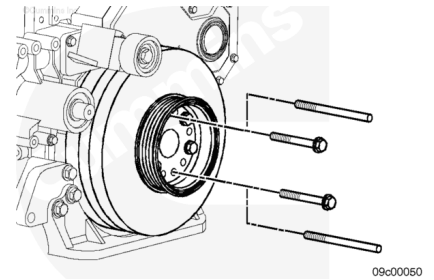
Remove the crankshaft adapter (1) from the pulley (2).



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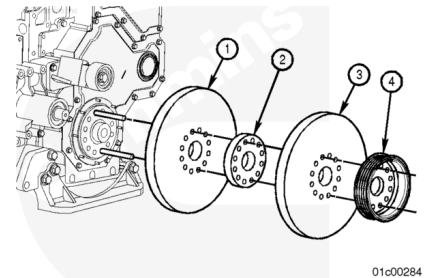
Remove two pulley capscrews.

Install two guide studs into the holes.



⚠ CAUTION ⚠

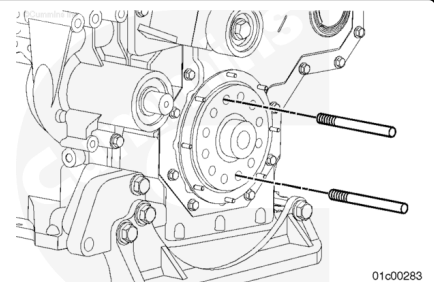
Do not use a hammer or a screwdriver to remove the vibration damper. These tools can damage the vibration damper.



Remove the remaining four capscrews.

Remove the pulley (4), vibration damper (3), mounting spacer (2) and second vibration damper (1).

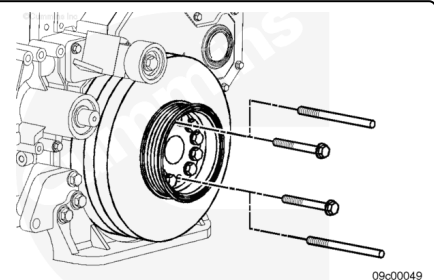
Remove the guide studs from the crankshaft nose.



Industrial and Power Generation without Crank-Driven Fan or PTO

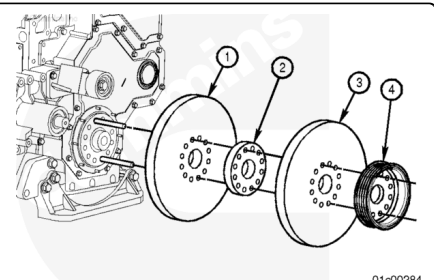
Remove two of the pulley capscrews.

Install two guide studs into the holes.



⚠ CAUTION ⚠

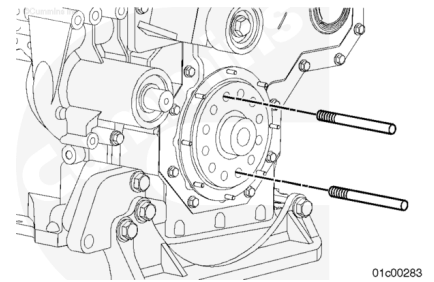
Do not use a hammer or a screwdriver to remove the vibration damper. These tools can damage the vibration damper.



Remove the remaining 10 capscrews.

Remove the pulley (4), vibration damper (3), mounting spacer (2), and second vibration damper (1).

Remove the guide studs from the crankshaft nose.

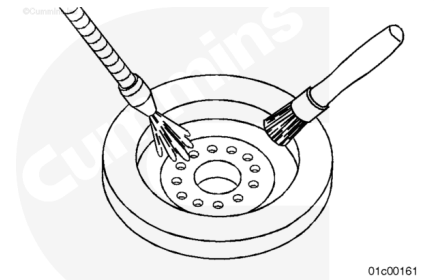


Clean and Inspect for Reuse

All Applications

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the vibration damper(s) with a solvent cleaner.

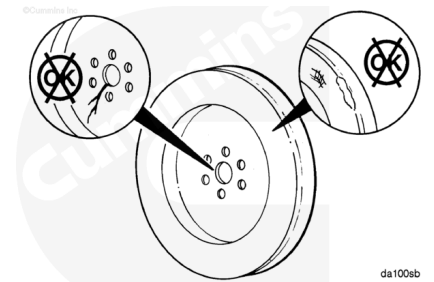
Always replace the vibration damper(s) after a crankshaft nose failure. Crankshaft nose breakages are primarily caused by excessive torsional activity.

The vibration damper **must always** be measured after a failure involving broken gears in the front gear train.

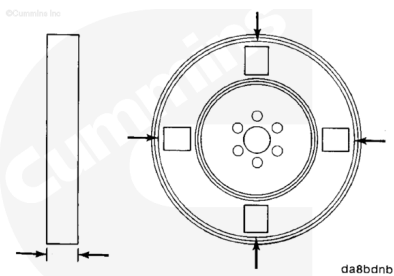
Check the mounting web for cracks.

Check the housing for dents or raised surfaces.

The vibration damper **must** be replaced if the mounting web is cracked or the housing has dents or raised surfaces.

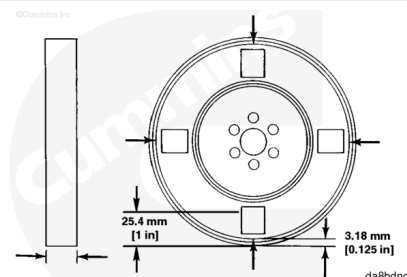


Remove the paint on both sides of the vibration damper in four locations spaced 90 degrees apart.



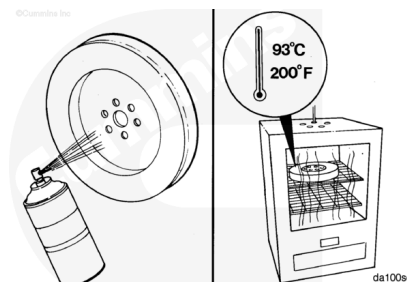
Measure and record the thickness at two points at each of the four locations.

The difference between any two of the eight measurements **must not** exceed 0.25 mm [0.010 in].



Spray the vibration damper with spot check developer, Type SKD-NF, or equivalent.

Heat the vibration damper in an oven (rolled lip side down) at 93°C [200°F] for 2 hours.

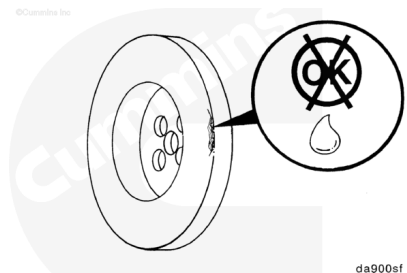


⚠ WARNING ⚠

Wear protective gloves when handling parts that have been heated to reduce the possibility of personal injury.

Remove the vibration damper from the oven and check for fluid leakage.

If there is leakage, replace the vibration damper.

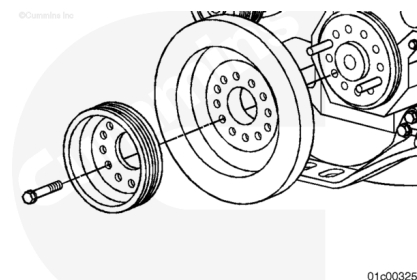


Install

Automotive Applications

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Make sure the mounting surfaces of the crankshaft nose, vibration damper, and the pulley are clean, dry, and free from burrs.

Install two guide studs into the crankshaft nose.

Align the dowel pin in the crankshaft with the dowel hole in the vibration damper.

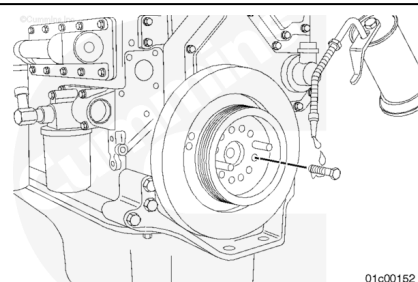
Install the vibration damper, pulley, and clamping plate onto the guide plate.

Use clean 15W-40 oil to lubricate the capscrew threads.

Install 10 capscrews.

Remove the two guide studs and install the remaining capscrews.

Tighten the capscrews in a star pattern.

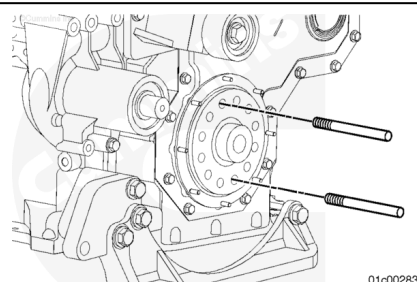


Torque Value: 175 n·m [130 ft-lb]

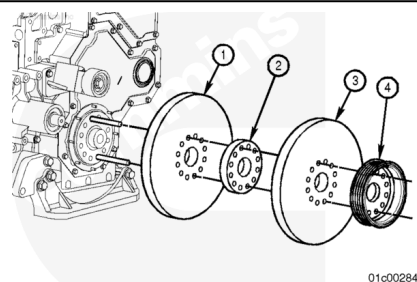
Industrial and Power Generation with Crank-Driven Fan or PTO

Make sure the mounting surfaces of the crankshaft nose, vibration dampers, mounting spacer, and pulley are clean, dry, and free from burrs.

Install two guide studs into the crankshaft nose.



Install one of the vibration dampers (1), mounting spacer (2), second vibration damper (3), and the pulley (4) onto the guide studs.



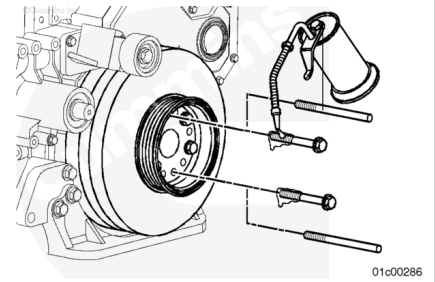
Lubricate the capscrew threads with clean 15W-40 oil.

Install four capscrews.

Remove the two guide studs and install the remaining two capscrews.

Tighten the six capscrews in a star pattern.

Torque Value: 175 n•m [130 ft-lb]



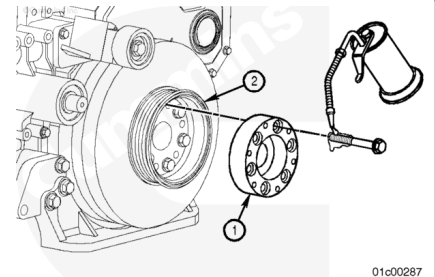
Install the crankshaft adapter (1).

Make sure the crank shaft adapter sets flush against the pulley (2).

Lubricate the capscrew threads with clean 15W-40 oil.

Install the six capscrews and tighten in a star pattern.

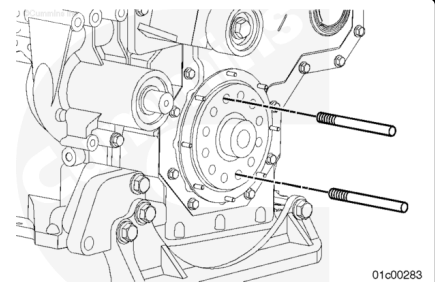
Torque Value: 175 n•m [130 ft-lb]



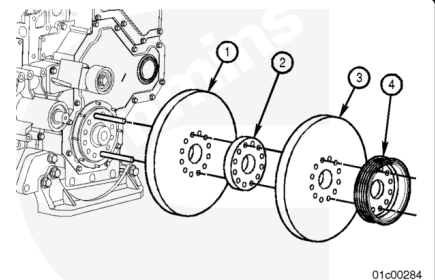
Industrial and Power Generation without Crank-Driven Fan or PTO

Make sure the mounting surfaces of the crankshaft nose, vibration dampers, mounting spacer, and pulley are clean, dry, and free from burrs.

Install two guide studs into the crankshaft nose.



Install one of the vibration dampers (1), mounting spacer (2), second vibration damper (3), and the pulley (4) onto the guide studs.



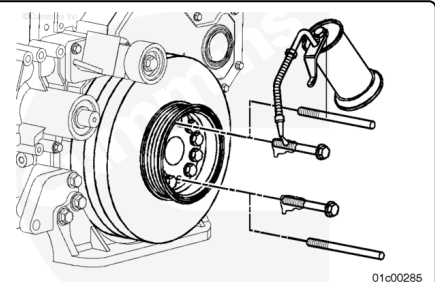
Lubricate the capscrew threads with clean 15W-40 oil.

Install 10 capscrews.

Remove the two guide studs and install the two remaining capscrews.

Tighten the 12 capscrews in a star pattern.

Torque Value: 175 n•m [130 ft-lb]



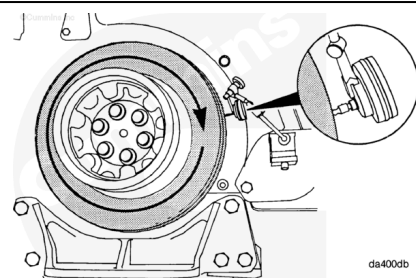
Eccentricity Check

All Applications

Clean the outside surface of the vibration damper.

To measure the vibration damper eccentricity (out of round), install a dial indicator onto the gear cover as indicated.

Rotate the crankshaft with the accessory driveshaft one complete revolution (360 degrees), and record the total indicator movement.



Vibration Damper Eccentricity		
mm		in
0.28	MAX	0.011

If the vibration damper is **not** within specifications, it **must** be replaced.

Wobble Check

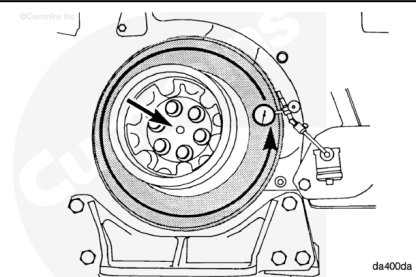
All Applications

To measure wobble (face alignment), install a dial indicator as shown.

Push the crankshaft to the front or rear of the engine and set the indicator to "0" (zero).

Rotate the crankshaft one complete revolution (360 degrees) while maintaining the position of the crankshaft, either toward the front or rear of the engine.

Record the total indicator movement.



Vibration Damper Wobble Measurement		
mm		in
0.28	MAX	0.011

If the vibration damper is **not** within specifications, it **must** be replaced.

Finishing Steps

Automotive Applications

- Install the fan and water pump belt tensioner. Refer to Procedure 008-080 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>).
- Install the fan and water pump drive belt. Refer to Procedure 008-002 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-002.html>).
- Install alternator drive belt. Refer to Procedure 013-005 in Section 13 (</qs3/pubsys2/xml/en/procedures/10/10-013-005.html>).
- Operate the engine and check for proper operation.



Industrial and Power Generation with Crank-Driven Fan or PTO

- Install the fan and water pump belt tensioner. Refer to Procedure 008-080 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>).
- Install the alternator drive belt. Refer to Procedure 013-005 in Section 13 (</qs3/pubsys2/xml/en/procedures/10/10-013-005.html>).
- Install the fan and water pump drive belt. Refer to Procedure 008-002 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-002.html>).
- Install the fan. Refer to the OEM service manual.
- Operate the engine and check for proper operation.



Industrial and Power Generation without Crank-Driven Fan or PTO

- Install the fan and water pump belt. Refer to Procedure 008-080 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>).
- Install the alternator drive belt. Refer to Procedure 013-005 in Section 13 (</qs3/pubsys2/xml/en/procedures/10/10-013-005.html>).
- Install the fan and water pump drive belt. Refer to Procedure 008-002 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-002.html>).



- Install the fan. See equipment manufacturer service information.
- Operate the engine and check for proper operation.

Last Modified: 17-Dec-2019
